If/Then Decisions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

If/Then Decisions

> Prepare If/Then Decisions: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Evaluate a **condition** using the current state and trace only the selected **branch**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Algorithms as Recipes

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

Look for: Trace an ordered algorithm using the current value after every instruction

Reference example, not the only acceptable wording: The trace is 3, 7, 14, 9. The doubling step uses 7, the current value after adding 4 to 3. It must not double the initial 3.

Accept a table, arrows or clearly labelled sentences that preserve the order.

Categories and Labels

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a category definition using explicit **evidence** and boundary rules

Reference example, not the only acceptable wording: In order: ready because both known fields pass; revise because 31 exceeds 30; review because the known day passes but the room is unknown; revise because the known day failure has priority over the unknown room.

Equality at 30 is accepted.

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Look for: Execute a stated rule including its boundary

Reference example, not the only acceptable wording: At least 5 gives narrow, wide, wide. Greater than 5 gives narrow, narrow, wide. Only the equality case changes among these **input**s.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

Concept to explain

Conditions make an algorithm **branch** when information changes.

Term	Meaning
condition	question evaluated as true or false
branch	path selected by a condition
threshold	value where a decision changes
else	path used when earlier conditions are false

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

This week combines current-state tracing with a single if/**else** decision. Use known whole-number scores first, then explicitly introduce missing **input**. Separate testing a **condition** from updating a value.

At least and at most include equality; above and below exclude it. Exactly one **branch** runs for each decision reached. A changed value does not cause the decision to repeat unless another test is explicitly specified.

All game and brightness examples are paper simulations.

Readiness answer

The addition changes score to 6, so the next instruction doubles 6 and produces 12. Each instruction reads the current state.

A score box, an add-2 card, and a paired if/otherwise instruction card with report and stop beneath it

A three-column trace headed Current value, Test result and Chosen update, plus cards 5, 6 and 7

The brightness-display task from W4 for independent work, and one unreadable-value card

End goal for the pictorial case

By the end, I can trace exactly one **branch** and test an including equality boundary.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model If/Then Decisions using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

2. Process — the deciding step

PROCESS check ordered if/then **condition**s

3. Output — the result

OUTPUT clothing suggestion and reason

Key idea

CORE IDEA **Condition**s make an algorithm **branch** when information changes.

How the mechanism works

UNDER THE HOOD A **condition**al evaluates a Boolean expression and selects a **branch**. Order matters when **condition**s overlap. Complete decision logic needs explicit handling for boundary values and an **else** path.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A clothing helper follows different paths depending on temperature, rain, and wind **condition**s.

Step 1

STEP 1 — OBSERVE weather values

Step 2

STEP 2 — TRACE check ordered if/then **condition**s

Step 3

STEP 3 — CONCLUDE clothing suggestion and reason

Model answer

MODEL REASONING Each test follows the stated **condition** order. The exactly-at-**threshold** case reveals whether the rule uses less than or less-than-or-equal. Rain-plus-cold needs combined **output**s or a clear priority.

A claim to avoid

DO NOT OVERCLAIM Never hide an unhandled case between two **threshold**s.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

If a card has a star, send it to A; otherwise send it to B.

The teaching case

Sort a star, a circle, and a card with both shapes.

Exact explanation to model

Why the result follows

Star and mixed cards go to A because both have a star. The circle goes to B. The test is has a star, not has only a star.

Explain the added worked example

Explain the added worked example · 1

A paper game updates a score with this procedure: add 2 to score; if score is at least 6, subtract 3 from score; otherwise add 4 to score; report score and stop. The if and otherwise instructions form one if/**else** decision.

Exactly one of those **branch**es runs each time this decision is reached.

A **condition** asks whether a statement is true for the current state. At least 6 means 6 or any larger number. If score is 6, the **condition** is true. If score is 5, it is false.

The **condition** itself does not change score; the chosen **branch** performs the update.

Start score at 4. First add 2, giving 6. Now test the current 6: the **condition** is true, so subtract 3 and report 3. Testing the original 4 would choose the wrong **branch**.

Write the state immediately before the **condition** so the **evidence** for the choice is visible.

Compare the three values immediately around the **threshold**. These are the values when the **condition** is reached, after the initial add-2 instruction.

Testing only a value far above 6 would not show whether equality was handled correctly.

Above 6 is different: it means strictly greater than 6, so 6 itself does not pass. At most 6 means 6 or smaller. Below 6 means strictly smaller, so it excludes 6. Say the equality case aloud before using one of these phrases.

A **condition** can be true even when its chosen action makes the number smaller. True describes the test result; it does not mean good, bigger, or a successful game. The written instructions define the action.

Key terms

Term	Meaning
Condition	A test whose result is true or false when the required information is known
Branch	The instruction or group of instructions selected by a condition
Threshold	The value used as a comparison boundary
Inclusive boundary	A boundary whose equality case is included
Boundary test	A test using a value at or close to a point where behaviour can change

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Initial score	State and condition after adding 2	Branch and final score
3	5 is not at least 6	Add 4 and report 9
4	6 is at least 6	Subtract 3 and report 3

Initial score	State and condition after adding 2	Branch and final score
5	7 is at least 6	Subtract 3 and report 4

Model the reasoning aloud · 2

In the first run, the false **branch** increases 5 to 9. That does not make the earlier decision run again. The procedure has already chosen and carried out the otherwise **branch**, so it moves on to report and stop.

Subtracting 3 afterward would incorrectly run both **branch**es and report 6.

This rule applies to one if/**else** decision. A later, separate **condition** would be a new test of the state at that later point. Do not invent an extra test or loop because the value crossed the **threshold**.

Follow only the instructions in the procedure.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: If/Then Decisions

A fictional counter begins with a number, adds 2, then chooses one **branch**.

The facts we will use

> **Supplied facts**
>
> If the updated value is at least 6, subtract 1; otherwise double it.

Supplied facts

Report once and stop. Use separate starting values 3,4,5.

AI ACADEMY / AI-1/10

Inclusive branching trace

	Start	After +2	At least 6?	Report
Run 1	3	5	false	10
Run 2	4	6	true	5
Run 3	5	7	true	6

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/10

If/Then Decisions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Update first

Starts 3,4,5 become 5,6,7.



2. Test current value

5>=6 is false; 6>=6 and 7>=6 are true.



3. Choose one branch

Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6.



4. Report

The three independent runs report 10,5,6.

CHANGE ONE THING

Replace at least 6 with above 6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Update first	Starts 3,4,5 become 5,6,7.
2. Test current value	5≥6 is false; 6≥6 and 7≥6 are true.
3. Choose one branch	Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6.
4. Report	The three independent runs report 10,5,6.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The **condition** is evaluated after adding 2. Equality belongs to the at-least **branch**, so starting at 4 checks the boundary.

Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

What this example does not establish

Keep the stated limit

A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

Change one thing in the pictured case

Change one thing

Change: Replace at least 6 with above 6.

Predicted result

Prediction: Only the run starting at 4 changes: it reports 12.

Why the result follows

Reason: Updated value 6 fails a strict above comparison and is doubled.

Ask while showing the picture

Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied **condition** and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate If/Then Decisions, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: If a card has a star, send it to A; otherwise send it to B.

2. State the decision to explain

Write the question to explain: Sort a star, a circle, and a card with both shapes.

3. Reveal the worked reasoning

Reveal after **prediction**s: Star and mixed cards go to A because both have a star. The circle goes to B. The test is has a star, not has only a star.

Hold back the independent task

Keep for an independent attempt: Use this exact rule: if a card has more than three dots, place it in tray A; otherwise use tray B. Trace cards with 2, 3 and 4 dots.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace **condition**s for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Choose initial score 4: add 2 to reach 6. Under at least 6, the test is true and subtracting 3 gives 3. Under above 6, it is false and adding 4 gives 10. Equality distinguishes the versions; test values 5 and 7 do not.

The exit answer is 6 at the **condition**, since it is included by at least but excluded by above. If testing from the start of the whole procedure, use initial 4 to reach that value.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the current-state rule from Week 9 using start 4, add 2, double. Record 4,6,12 before introducing a condition.
7 to 20 minutes	Model the paper-game procedure from initial score 4. Mark the value 6 at the test and show that the test itself changes no value.
20 to 33 minutes	Compare values 5,6,7 at the condition. Say equality aloud for at least, above, at most and below. Keep separate runs visually separate.
33 to 45 minutes	Complete W1 and W2 in pairs. Require a true/false result and chosen branch alongside each final value.
Session 2 start 0 to 8 minutes	Trace initial score 3 to current 5, then false branch to 9. Point to report and stop so crossing the threshold does not trigger another branch.

Stage	Teacher action
8 to 23 minutes	Use W3 to diagnose running both branches. Complete the guided equality test and compare expected versus copied instructions.
23 to 38 minutes	Collect W4 independently. Check whether students evaluate at most on 6,7,8 rather than on the starting 3,4,5.
38 to 45 minutes	Use W5 to distinguish unknown from false. Discuss the exit test and preview explicitly repeated tests in loops.

Reduce help deliberately

Completion practice

Completion task: The known value at a **condition** is exactly 6. Decide whether each test is true or false: at least 6; above 6; at most 6; below 6. Explain which phrases include equality.

Then name one value below and one value above 6 for neighbouring tests.

Reasoning to listen for

Teacher check: At least 6 is true, above 6 is false, at most 6 is true, and below 6 is false. At least and at most include equality; above and below are strict.

The nearest whole-number neighbours are 5 and 7; other correctly identified lower and higher values are valid but are less close to the boundary.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Condition**s and boundary cases. Proposed explanation: “The **condition** reads the starting value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: Trace the initial add-2 instruction first and circle its result before evaluating the comparison.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which **input** detects the difference between the two written **condition**s? Explain its entire run.

Reasoned teacher answer

Start 4: add 2 gives 6. The including equality rule subtracts 1 to report 5; the strict rule doubles to report 12.

Changed-case reasoning: Only the run starting at 4 changes: it reports 12. Updated value 6 fails a strict above comparison and is doubled.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the If/Then Decisions investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Build a weather-clothing rule and test four cases.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

> **Find the unsupported claim**
>
> CLAIM TO REPAIR More than three includes three.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The **condition** reads the starting value: Trace the initial add-2 instruction first and circle its result before evaluating the comparison.

At least means strictly above: Put 6 on the boundary and read at least as 6 or more. Compare its true result with above 6, which is false.

Crossing the **threshold** runs the other **branch**: Follow the chosen **branch** to report and stop. A second test must be explicitly written; it is not implied by the new value.

An unknown value makes the test false: Show that the unreadable value could be 5 or 6. Different possibilities select different **branch**es, so the exercise requires a pause and clarification.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the If/Then Decisions questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Star and mixed cards go to A because both have a star. The circle goes to B. The test is has

a star, not has only a star.

Independent question

Independent question · withhold the key

Use this exact rule: if a card has more than three dots, place it in tray A; otherwise use tray B. Trace cards with 2, 3 and 4 dots.

Expected independent answer

Teacher key · Expected independent answer

Two and three dots go to B; four goes to A. More than three excludes three. The boundary case reveals whether the **condition** was read precisely.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification **threshold**.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “More than three includes three.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

With AND only ticket=yes/adult=yes is admitted. With OR the two mixed cases are also admitted; no/no is still refused. AND requires both **condition**s, whereas OR requires at least one.

The policy must be chosen before judging the code.

Explanations for the additional workbook examples

W1 Trace a decision from the current score

Workbook question · W1

Use the paper-game procedure: add 2 to score; if score is at least 6 subtract 3, otherwise add 4; report and stop. Start at 2, then make a separate run starting at 5.

For each run record the value at the **condition**, its true or false result, the **branch** and the final score.

Teacher reasoning · W1

Start 2: after adding 2 score is 4; at least 6 is false; add 4 and report 8. Start 5: after adding 2 score is 7; at least 6 is true; subtract 3 and report 4. Each run starts fresh and executes only its selected **branch**.

W2 Read four boundary phrases

Workbook question · W2

The known value at a **condition** is exactly 6. Decide whether each test is true or false: at least 6; above 6; at most 6; below 6. Explain which phrases include equality.

Then name one value below and one value above 6 for neighbouring tests.

Teacher reasoning · W2

At least 6 is true, above 6 is false, at most 6 is true, and below 6 is false. At least and at most include equality; above and below are strict.

The nearest whole-number neighbours are 5 and 7; other correctly identified lower and higher values are valid but are less close to the boundary.

W3 Stop after the selected branch

Workbook question · W3

In a separate decision the current score is 4. If score is at least 5 subtract 2; otherwise add 3; report and stop. A learner adds 3 to get 7, then subtracts 2 because 7 is now at least 5, and reports 5.

Identify the extra action and give the correct trace and result.

Teacher reasoning · W3

The **condition** at score 4 is false, so only the otherwise **branch** runs. Add 3 to get 7, then report 7 and stop. The extra subtraction incorrectly runs the other **branch** after the decision.

Crossing the **threshold** does not repeat the test in this if/**else** procedure.

W4 Transfer the method to a new display

Workbook question · W4

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5.

Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

Teacher reasoning · W4

Start 3: 6 at the test, true, subtract 2, report 4. Start 4: 7 at the test, true, subtract 2, report 5. Start 5: 8 at the test, false, add 1, report 9. Starting at 4 tests equality.

With below 7, the current 7 fails, so add 1 and report 8 instead of 5. All three runs use current values after adding 3.

W5 Handle an unknown input explicitly

Workbook question · W5

The score is unreadable at the at-least-6 test. A helper says the test must be false and takes the otherwise **branch**. Explain what is unknown and why that choice is unjustified.

State the next step required by this paper exercise and give two possible scores that would select different **branch**es.

Teacher reasoning · W5

The score value, and therefore the truth of the comparison, is unknown. It could be 5, which selects otherwise, or 6, which selects the true **branch**. Neither is established by an unreadable value.

Pause and request the missing score as specified; do not silently replace it with zero or treat the test as false.

Assess independent transfer

Independent question · withhold the key

Selected case: A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5.

Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

Selected case answer

Teacher key · Selected case answer

Start 3: 6 at the test, true, subtract 2, report 4. Start 4: 7 at the test, true, subtract 2, report 5. Start 5: 8 at the test, false, add 1, report 9. Starting at 4 tests equality.

With below 7, the current 7 fails, so add 1 and report 8 instead of 5. All three runs use current values after adding 3.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect If/Then Decisions to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Add one transparent if/then **branch** and one **else** path to the Classroom Sorter.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 11: Repeating Efficiently. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a number line with the boundary labelled and let students read each comparison in words. Use text labels True **branch** and Otherwise **branch** in addition to colour. Supply a blank trace with the comparison row highlighted.

Extension: explain why checking 5,6,7 is useful but does not prove a program correct for every possible **input** or every unrelated implementation error.

W4 is secure when the learner traces all three runs, applies the test to the updated value, includes equality for at most, and explains the changed **output** under below.

A final answer without a **branch** reason is partial **evidence**. After support, retry with add 2 followed by if current value is at most 5 subtract 1, otherwise add 2.

From starts 2,3,4, the correct final values are 3,4,8 respectively.

Week 11 introduces loops and stopping rules. Students will distinguish one decision from an explicitly repeated test and trace when a loop stops.

Plan the delayed return

Suggested return points in this level: ai-1/11, ai-1/13, ai-1/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**.

Prepare, teach and assess

> Use this preparation summary alongside the detailed If/Then Decisions lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Evaluate a **condition** using the current state and trace only the selected **branch**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: The known value at a **condition** is exactly 6. Decide whether each test is true or false: at least 6; above 6; at most 6; below 6. Explain which phrases include equality.

Then name one value below and one value above 6 for neighbouring tests.

Supported practice answer

Reasoned practice answer: At least 6 is true, above 6 is false, at most 6 is true, and below 6 is false. At least and at most include equality; above and below are strict.

The nearest whole-number neighbours are 5 and 7; other correctly identified lower and higher values are valid but are less close to the boundary.

Check every learner before discussion: Use the worked case for **Condition**s and boundary cases. Proposed explanation: “The **condition** reads the starting value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Trace the initial add-2 instruction first and circle its result before evaluating the comparison.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Replace at least 6 with above 6. Require a **prediction** and a reason before comparing. Expected result: Only the run starting at 4 changes: it reports 12.

Reason: Updated value 6 fails a strict above comparison and is doubled.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The **branch**-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

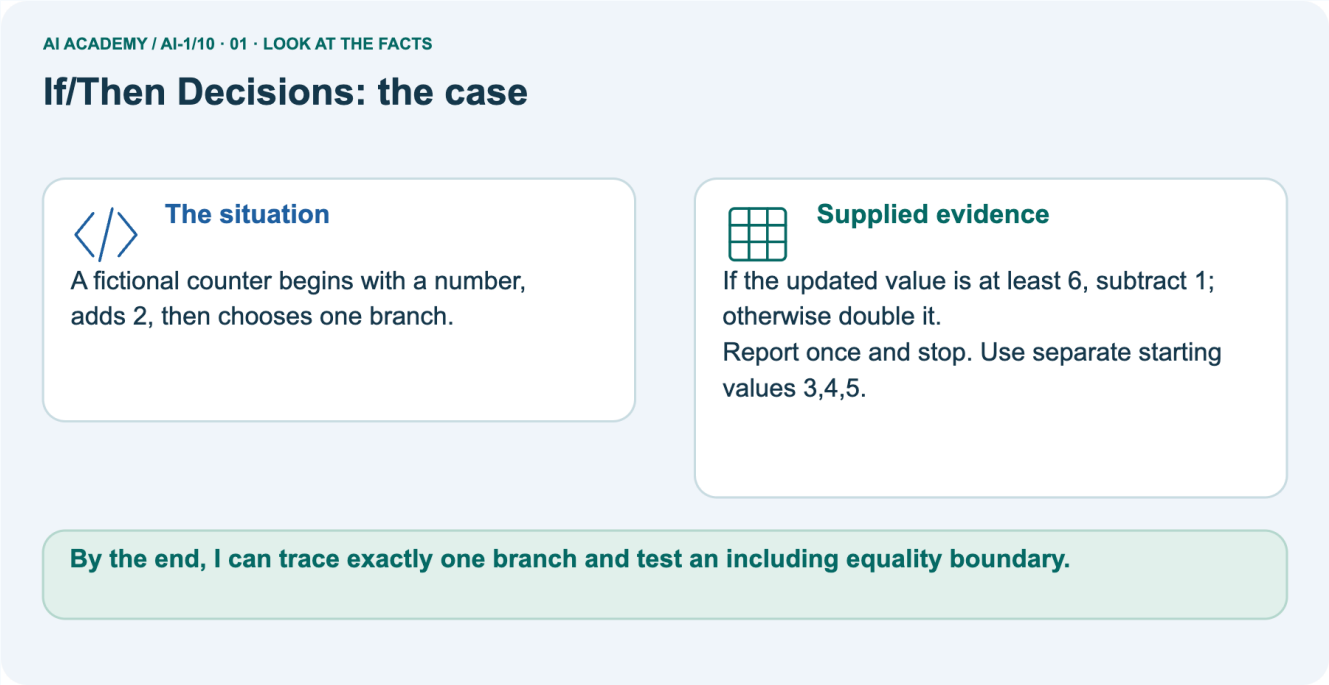
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one If/Then Decisions graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. If/Then Decisions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

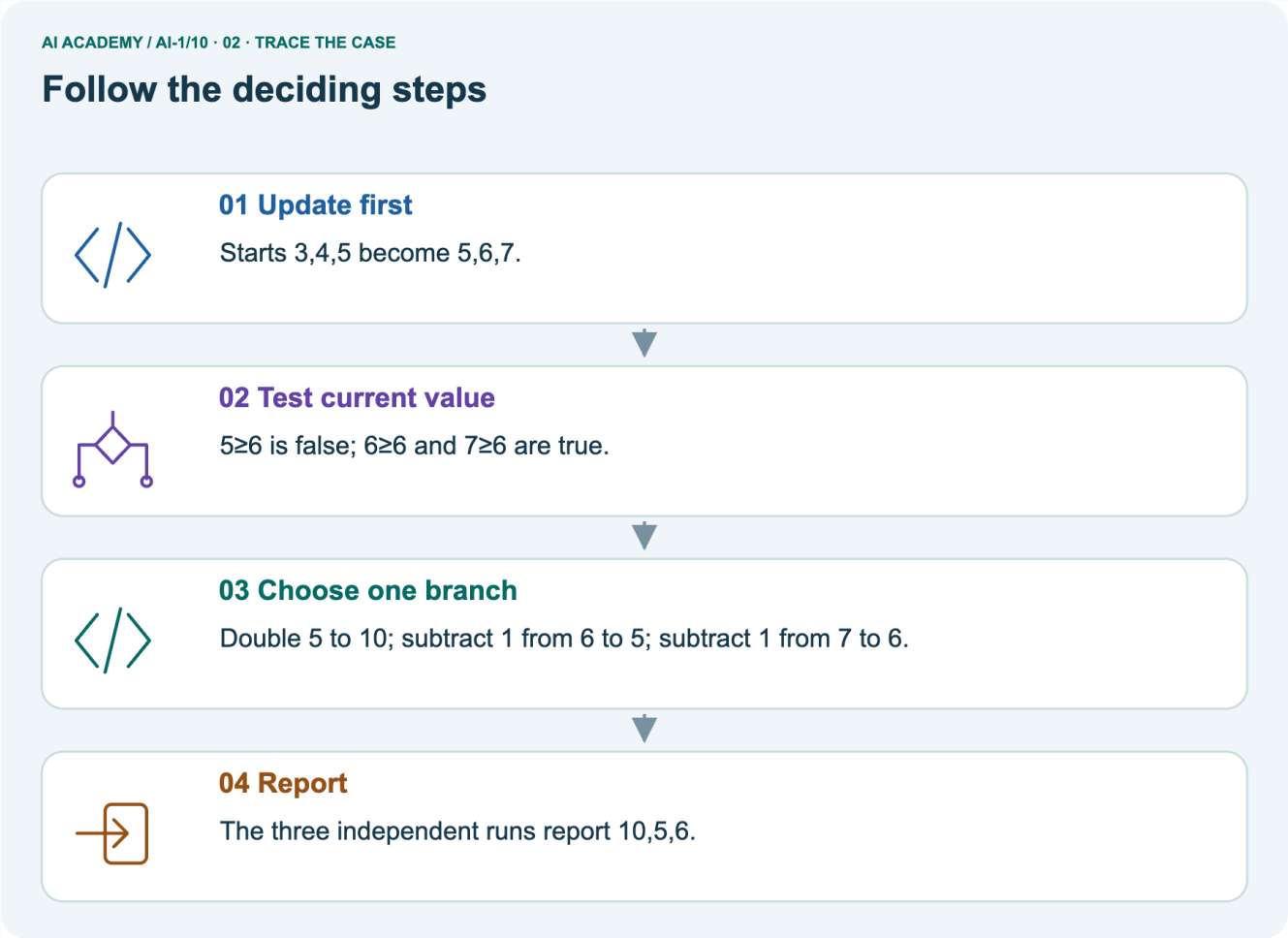
A fictional counter begins with a number, adds 2, then chooses one **branch**. Supplied facts: If the updated value is at least 6, subtract 1; otherwise double it. Report once and stop. Use separate starting values 3,4,5.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

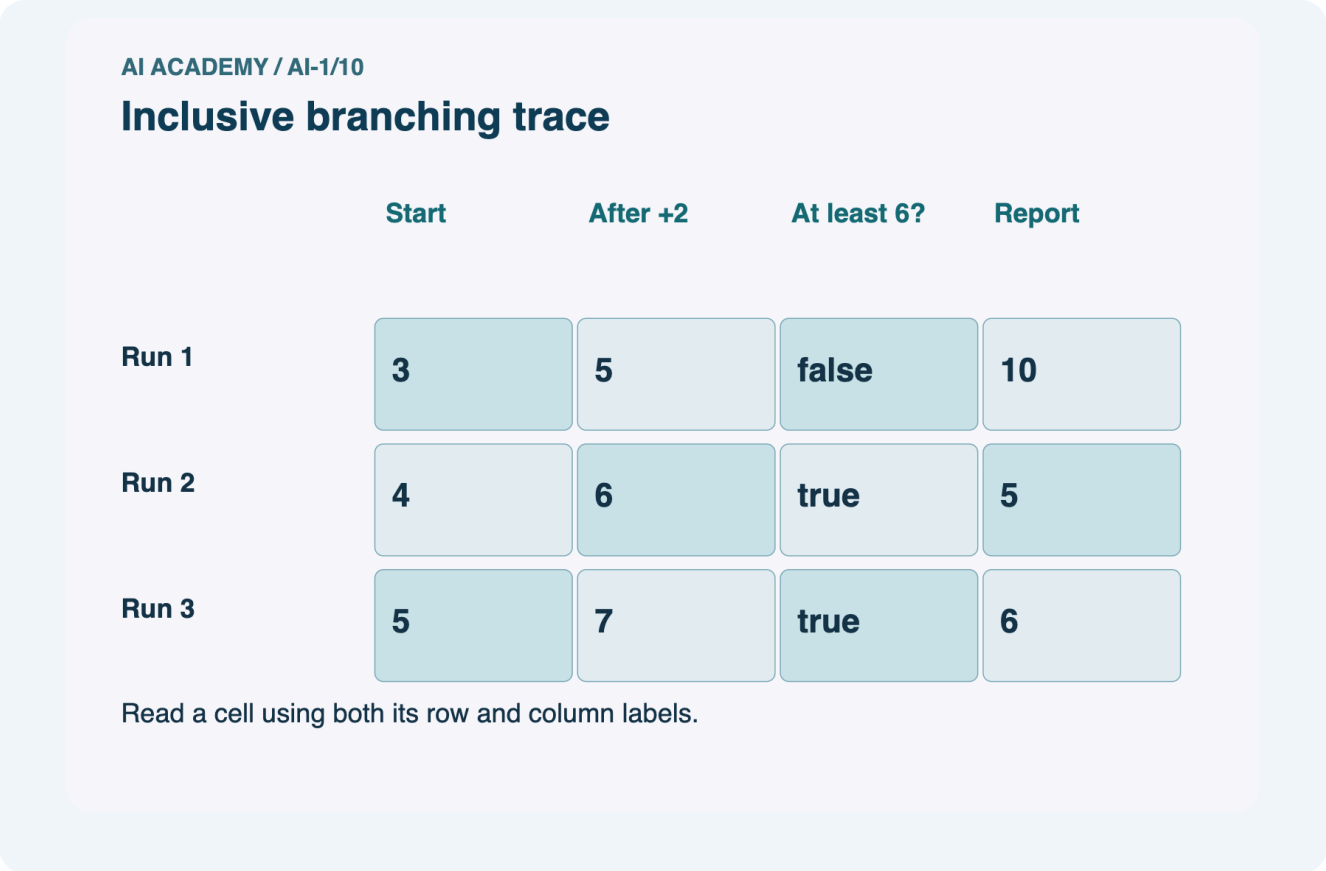
Update first: Starts 3,4,5 become 5,6,7. → Test current value: 5≥6 is false; 6≥6 and 7≥6 are true. → Choose one **branch**: Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6. → Report: The three independent runs report 10,5,6.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Inclusive branching trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Inclusive **branch**ing trace. Run 1: Start = 3, After +2 = 5, At least 6? = false, Report = 10. Run 2: Start = 4, After +2 = 6, At least 6? = true, Report = 5. Run 3: Start = 5, After +2 = 7, At least 6? = true, Report = 6.

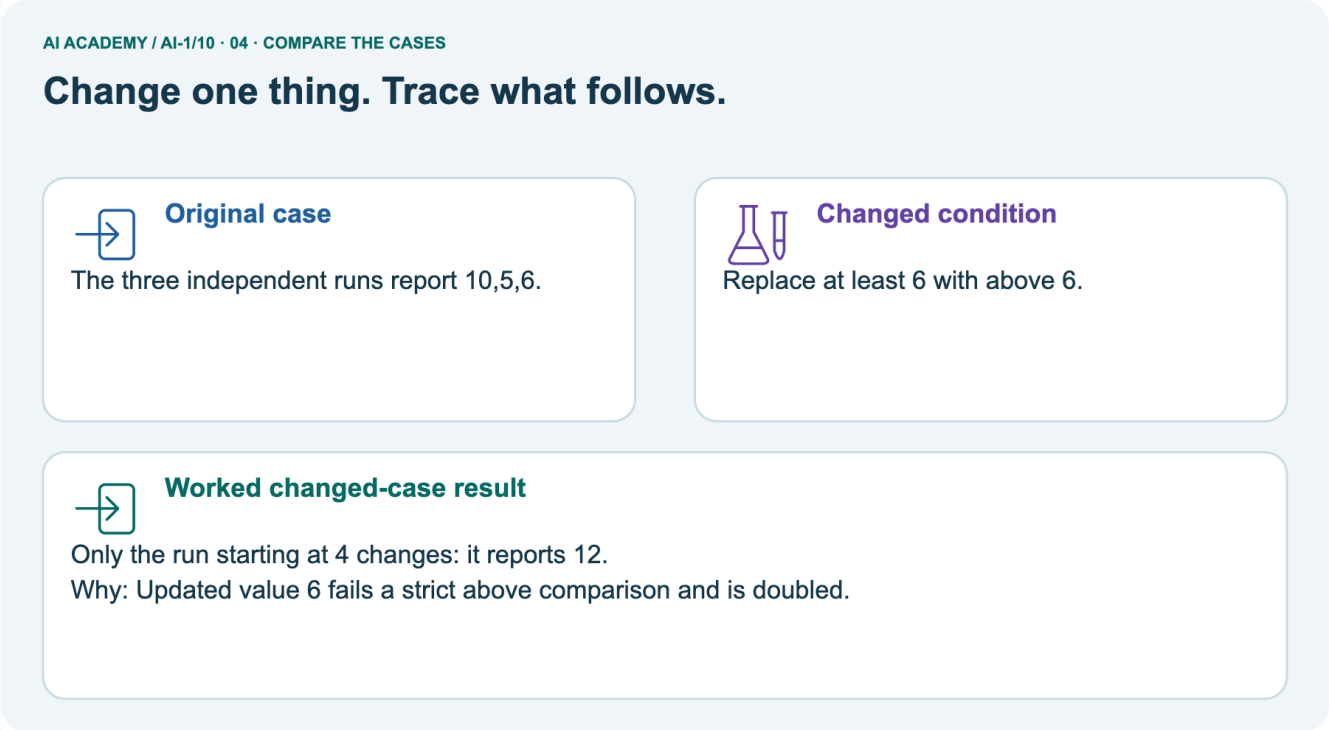
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The three independent runs report 10,5,6. Changed **condition**: Replace at least 6 with above 6. Expected result: Only the run starting at 4 changes: it reports 12.

Reason: Updated value 6 fails a strict above comparison and is doubled.

Predict before checking the worked changed-case result. Name what changed and what the

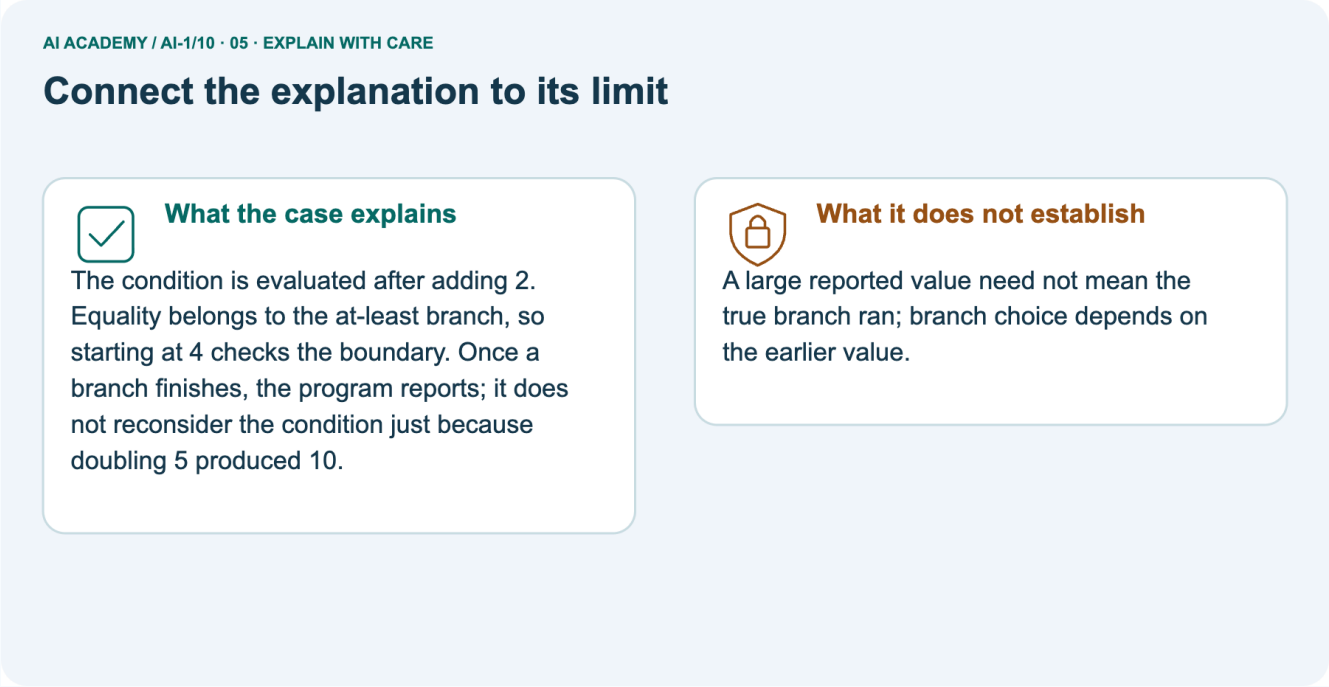
supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Updated value 6 fails a strict above comparison and is doubled.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **condition** is evaluated after adding 2. Equality belongs to the at-least **branch**, so starting at 4 checks the boundary.

Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Limit: A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

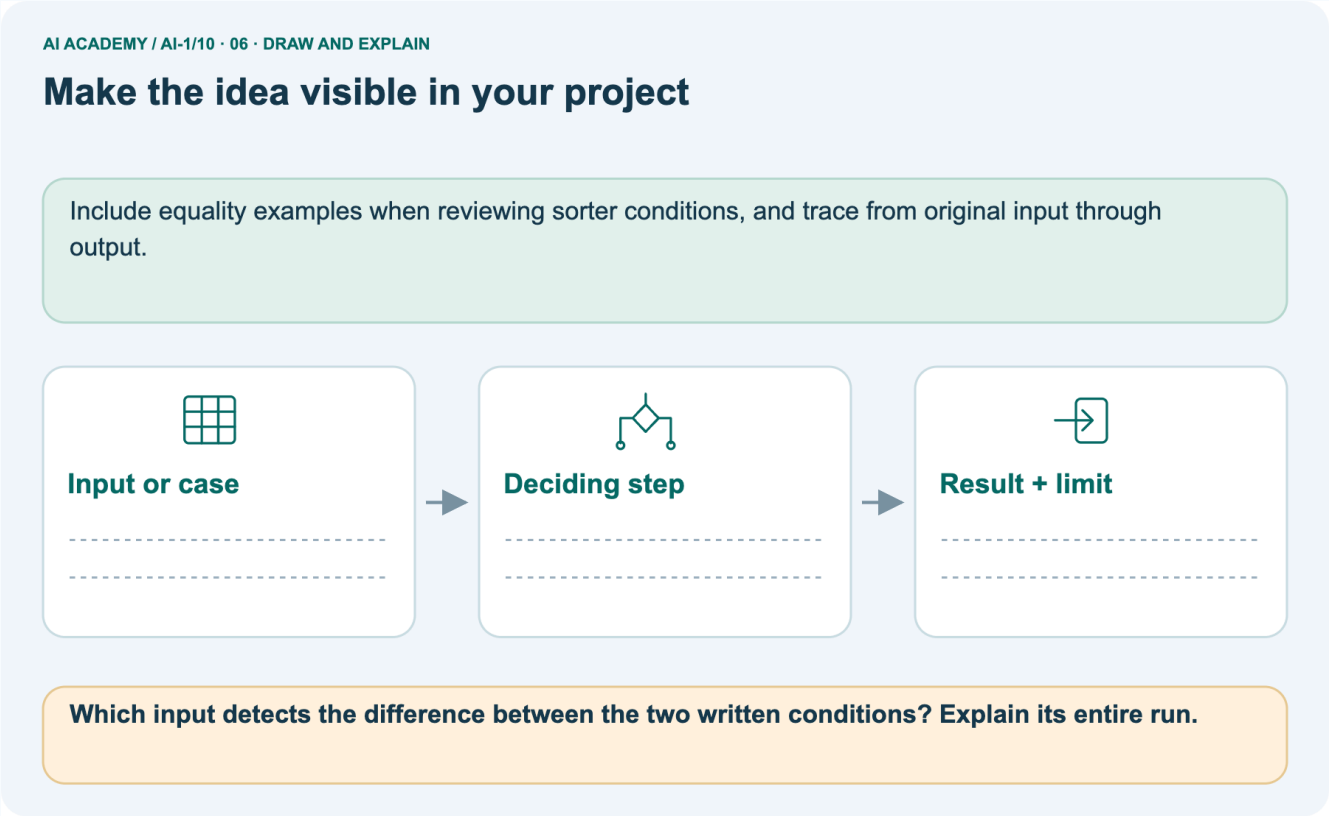
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** detects the difference between the two written **condition**s? Explain its entire run.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Start 4: add 2 gives 6. The including equality rule subtracts 1 to report 5; the strict rule doubles to report 12.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.